Rayen Ghali

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Education

University of Moncton, Faculty of Engineering — **May 2026**
Master of Applied Science (MASc) | GPA: 4.18/4.3 — *New Brunswick, Canada*

University of Carthage, National Institute of Applied Science and Technology (INSAT) — **Sept 2023**
Industrial and Systems Engineering | Graduated with High Honours — *Tunis, Tunisia*

Projects

End-to-End Multimodal AI Agent for Autonomous Robotic Control

Master's Thesis

- Designed an LLM-based agent integrated with a Vision-Language-Action (VLA) policy for end-to-end robotic control, enabling autonomous task planning, error recovery, and real-time action execution from visual observations
- Applied parameter-efficient fine-tuning and reinforcement learning to adapt vision-language models for robotic action generation
- Established evaluation benchmarks measuring success rates and latency across VLA architectures on real-world manipulation tasks and simulation environments
- Orchestrated distributed multi-GPU training on Compute Canada's CCDB cluster via Slurm, using W&B for experiment tracking and reproducibility

ChunkCanvas

- Built a full-stack document-processing application for RAG workflows, supporting multimodal parsing of PDFs, images, and audio/video using vLLM for local model serving alongside cloud inference engines, LangChain-based chunking, embedding generation, and vector database ingestion

Experience

Applied AI Research Assistant — **Apr 2023 – Present**
LARIHS, University of Moncton — *New Brunswick, Canada*

LLM-Driven Industrial Robot Control

- Engineered a multimodal agent using LangGraph to control an industrial KUKA robot via natural language (text and speech)
- Integrated persistent conversational memory and a toolset for pick-and-place, Cartesian, and joint space movements, achieving <2.6s end-to-end latency
- Implemented a robust error recovery system featuring safety checks, rollback, automatic retries, and real-time voice feedback
- Co-authored "LLM-driven agent for speech-enabled control of industrial robots: A case study in snow-crab quality inspection," published in *Results in Engineering*, 2025

Survey Data Analytics Platform

- Migrated 15 years of Acadian student survey data from 7 schools, converting Excel files to a centralized PostgreSQL database
- Built a multi-agent workflow using LangGraph with human-in-the-loop validation for natural language queries
- Developed a Streamlit web application with automated plotting for query results (chatcapitalhumain.ca); validated with research personnel and adopted to streamline survey data analysis

Genealogical RAG Conversational Agent

- Developed a RAG-based conversational agent for querying historical Acadian genealogical records (1700-1900), reducing average archivist search time from ∼15 minutes per query to under 30 seconds (chatacadien.ca)
- Engineered entity-centric chunking and context grounding solutions to disambiguate individuals with identical names in unstructured genealogical text, mitigating a primary cause of model hallucinations
- Evaluated and validated the system for faithfulness, context precision, and recall on a curated dataset of real user interactions
- Built complementary web application for archival image search with VLM-generated descriptions (chatpatrimoineacadien.ca)
- Co-authored and presented "ChatAcadien: A RAG-LLM-Based Chatbot for Exploring Acadian Genealogy" (IEEE CASCON 2025)

Industrial Defect Detection & Few-Shot Learning

- Benchmarked modern object detection models on industrial surface-defect datasets, evaluating accuracy and speed tradeoffs
- Developed SSL-YOLO, a few-shot learning framework combining contrastive self-supervised pre-training with YOLOv8
- Co-authored and presented "Real-time defect detection systems for steel and wood inspection" (IEEE CECCE 2024) and "Benchmarking few-shot learning techniques for steel surface defect detection" (IEEE SWC 2025)

Skills

Languages: Python, SQL
Technologies: PyTorch, Transformers, LangChain, FastAPI, vLLM, W&B, Supabase, Git, Docker, GCP

Languages

Arabic: Native | **French:** Bilingual | **English:** Proficient